MZmine Parameter Optimisation

A step-by-step explanation of the two-stage search and the Phase 1 and Phase 2 scoring formulas

What the script does
It performs a two-stage random parameter search. Phase 1 screens candidate parameter sets using one QC file. Phase 2 validates the best Phase 1 candidates across all QC files. The parameter set with the highest Phase 2 score is written into the final MZmine batch.

Workflow at a glance

1. Read configuration	2. Build full parameter grid	3. Randomly sample Phase 1 runs	4. Run one QC and score	5. Keep top candidates	6. Run all QCs and score	7. Select highest Phase 2 score

Optimised parameters

The grid contains candidate values for MS1 noise, minimum consecutive scans, minimum consecutive intensity, chromatogram-builder minimum absolute height, chromatographic threshold, resolver minimum height, peak top/edge ratio, resolver minimum scans, and minimum and maximum peak duration.

This is **random search with two-stage screening**, not gradient-based or Bayesian optimisation. Only combinations explicitly present in the configuration can be selected.

Core idea: Phase 1 asks whether a setting detects an informative feature list. Phase 2 asks whether the detected features are reproducible across QC runs.

1. Building and sampling the parameter space

The script uses **tidyr::crossing()** to generate the Cartesian product of all candidate values. If parameter i has n_i candidate values, the total number of combinations is:

$$N_{grid} = n_1 \times n_2 \times \ldots \times n_{10}$$

The complete grid may be very large, so Phase 1 samples at most **n_phase1_runs** combinations. The sample size is the smaller of the requested number of runs and the number of available combinations.

> **Reproducibility:** set.seed(random_seed) makes the same random parameter combinations selectable again when the configuration and grid remain unchanged.

Phase 1 execution

• Each sampled parameter set is written into a copy of the base MZmine batch.

• The imported data are replaced with qc_single.

• The retained modules are Import MS data, Mass detection, Chromatogram builder, Smoothing, Local minimum feature resolver, CSV export, and batch saving.

• MZmine is executed, the feature CSV is read, and a Phase 1 score is calculated.

• Successful runs are ranked by descending Phase 1 score; only top_n_phase2 candidates continue.

Useful interpretation

Phase 1 is a relatively inexpensive screen. It reduces the number of candidates before the more demanding all-QC validation.

2. Phase 1 scoring formula

Phase 1 evaluates feature detection using one QC file.

Phase 1 score

$$S_1 = \log_{10}(N + 1) + 10F_{MS2} + 0.5 \log_{10}(H_{median} + 1) - 0.0005N$$

Term	Meaning	Role in the score
N	Number of detected features	Log reward plus a linear penalty
FMS2	Features with MS2 / all features	Strong reward; contribution from 0 to 10
Hmedian	Median feature height	Modest log-scaled reward
-0.0005N	Feature-count penalty	Discourages overly permissive detection

How each term behaves

• **Feature reward:** $\log_{10}(N+1)$ rises slowly. Approximate values are 2 for 100 features, 3 for 1,000, and 4 for 10,000.

• **MS2 reward:** $10F_{MS2}$. An MS2 fraction of 0.30 contributes 3 points; 0.80 contributes 8 points.

• **Height reward:** $0.5\log_{10}(H_{median}+1)$ reduces the influence of very large intensity scales.

• **Feature penalty:** -0.0005N contributes -0.5 at 1,000 features, -2.5 at 5,000, and -5 at 10,000.

Interpretation: Phase 1 rewards informative, stronger features with MS2 support, while preventing raw feature count from becoming the only objective.

3. Phase 2 metrics and score

Phase 2 reruns the best Phase 1 candidates using all files in qc_all. It retains additional isotope-processing, alignment, and feature-list filtering modules, then evaluates cross-QC reproducibility.

How the reproducibility metrics are calculated

- Numeric abundance columns are identified after excluding metadata-like columns.
- Missing abundance values are replaced with zero.
- For each feature, detection frequency is the fraction of abundance columns with a value greater than zero.

$$D_i = \text{number of QC abundance values} > 0 \,/\, \text{number of abundance columns}$$

- For each feature, relative standard deviation is calculated as standard deviation divided by mean.

$$RSD_i = SD_i \,/\, mean_i$$

- The script then uses the median detection frequency and median RSD across features.

Phase 2 score

$$S_2 = 50D + 20F_{MS2} - 10R + 0.5 \log_{10}(H_{median} + 1) - 0.0005N$$

D is median detection frequency and R is median QC RSD.

4. Interpreting the Phase 2 terms

Component	Example	Score effect
50D	D = 0.2 / 0.5 / 0.8 / 1.0	+10 / +25 / +40 / +50
20FMS2	FMS2 = 0.10 / 0.25 / 0.50 / 1.0	+2 / +5 / +10 / +20
-10R	R = 0.10 / 0.20 / 0.50 / 1.0	-1 / -2 / -5 / -10
Height term	Median height increases by orders of magnitude	A modest positive contribution
-0.0005N	N = 1,000 / 5,000 / 10,000	-0.5 / -2.5 / -5

Relative importance

• **Detection frequency dominates:** its maximum contribution is 50 points, making consistent cross-QC detection the strongest explicit objective.

• **MS2 remains important:** its maximum contribution is 20 points.

• **QC variability is penalised:** larger median RSD reduces the score.

• **Intensity is secondary:** the logarithmic height term provides a smaller reward.

• **Excess feature count remains penalised:** the same linear penalty used in Phase 1 is retained.

Important implementation detail: replacing missing abundance values with zero affects both detection frequency and RSD. A feature missing in several QCs may therefore receive a lower detection-frequency contribution and a larger variability penalty.

Fallback behaviour

If the script cannot identify abundance columns, Phase 2 falls back to a Phase-1-like formula: log-scaled feature count + 10 x MS2 fraction + log-scaled median height - feature-count penalty.

5. Final selection and practical meaning

After all Phase 2 runs finish, the script removes non-finite scores, sorts candidates in descending order, and selects the first row. This complete parameter combination becomes the final set.

The winning parameters are then applied to:

• the final optimised full MZmine batch, optionally using final_input_files; and

• a QC-only validation batch using qc_all.

What “optimal” means here

Optimal means the tested parameter combination with the highest user-defined Phase 2 score. It does not mean a universal optimum. The result depends on the candidate values, random Phase 1 sample, QC files, abundance-column detection, missing-value handling, and formula weights.

Summary comparison

Aspect	Phase 1	Phase 2
Data	One QC file	All QC files
Purpose	Fast screening of detection quality	Validation of cross-QC reproducibility
Main rewards	Features, MS2 fraction, median height	Detection frequency, MS2 fraction, median height
Main penalties	Excess feature count	QC RSD and excess feature count
Output	Top candidates for Phase 2	Single winning parameter set

Source and scope

Prepared from the supplied mzmine_optimiser.txt script. Formula descriptions and workflow details reflect the implementation in that file. Any judgement about whether the weights are scientifically appropriate should be validated against the goals and characteristics of the experiment.

End of guide